(1)

Dated:

Given Dataset:-

S.No.	Outlook	Temperature	Humidity	windy	Rain(Target).
1	Sunny	Hot	High	weak	No
2	Sunny	Mild	High	Strong	No
3	Cloudy	Hot	High	weak	Yes
4	Rainy	Mild	High	weak	Yes
5	Rainy	Cool	Normal	weak	Yes
6	Rainy	Cool	Normal	Strong	NO
7	Cloudy	Mild	Normal	Strong	Yes
8	Sunny	Cool	Normal	weak	NO

Total Entropy of Dataset (S)

Total: 8, Yes = 4 , NO = 4

$$\text{Entropy}(S) = -\frac{4}{8}\log_2\frac{4}{8} - \frac{4}{8}\log_2\frac{4}{8}$$

$$= -\frac{1}{2}\log_2\frac{1}{2} - \log_2\frac{1}{2}$$

Entropy(S) = 1

Information Gain for each attribute:

Outlook:-

Outlook	Total	Yes	NO	Entropy
Sunny	3	0	3	0
Cloudy	2	2	0	0
Rainy	3	2	1	0.983

weighted Entropy:- $\frac{3}{8}(0) + \frac{2}{8}(0) + \frac{3}{8}(0.9183)$

$= 0.3444.$

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Dated:

IG(Outlook) = 1 - 0.3444 = 0.6556

Temperature :-

Temperature	Total	Yes	NO	Entropy
Hot	2	1	1	1
Mild	3	2	1	0.9183
Cool	3	1	2	0.9183

Weighted Entropy = $\frac{2}{8}(1) + \frac{3}{8}(0.9183) + \frac{3}{8}(0.9183$

= 0.9387

IG(Temperature) = 1 - 0.9387 = 0.613

Humidity :-

Humidity	Total	Yes	NO	Entropy
High	4	2	2	1
Normal	4	2	2	1

weighted Entropy = $\frac{4}{8}(1) + \frac{4}{8}(1) = 1$

IG(Humidity) = 1 - 1 = 0

Windy :

Windy	Total	Yes	NO	Entropy
Weak	5	3	2	0.9710
Strong	3	1	2	0.9183

weighted Entropy = $\frac{5}{8}(0.9710) + \frac{3}{8}(0.9183)$

IG(Windy) = 1 - 0.9512 = 0.0488

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(3)

Dated: _____

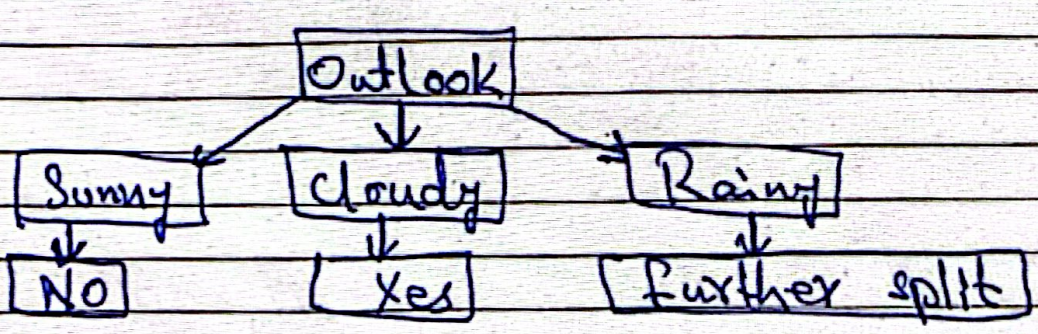
Summary of Information Grain:

Attribute	Information Grain
Outlook	0.6556
Temperature	0.0613
Humidity	0
Windy	0.0488

First Root Node of Decision Tree:

The attribute with heighest Information Grain is
Outlook = (0.6556).
So, the first rootd node of the decision tree is Outlook.

First Level of Decision Tree:-



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